Digital Electronic Circuits (EC2L004)

Lecture 2: Binary Numbers – Part II

Learning Objectives:

- Subtraction with Complements
- Signed Binary Numbers

Subtraction with Complements

The subtraction of two n-digit unsigned numbers M - N in base r can be done as follows:

1. Add the minuend M to the r's complement of the subtrahend N. Mathematically, $M + (r^n - N) = M - N + r^n$.

2. If $M \geq N$, the sum will produce an end carry r^n, which can be discarded; what is left is the result $M - N$.

3. If $M < N$, the sum does not produce an end carry and is equal to $r^n - (N - M)$, which is the r's complement of $(N - M)$. To obtain the answer in a familiar form, take the r's complement of the sum and place a negative sign in front.

Example: 1

- Using 10's complement, subtract 72532 – 3250.

$$
\begin{array}{rr}
M = & 72532 \\
\text{10's complement of} \quad N = & \underline{+96750} \\
\text{Sum} = & 169282 \\
\text{Discard end carry } 10^5 = & \underline{-100000} \\
\text{Answer} = & 69282
\end{array}
$$

Example: 2

- Using 10's complement, subtract 3250 - 72532.

$$
\begin{array}{rrr}
 & M = & 03250 \\
\text{10's complement of} & N = & \underline{+\,27468} \\
 & \text{Sum} = & 30718
\end{array}
$$

There is no end carry. Therefore, the answer is

- (10's complement of 30718) = - 69282

Example 3:

Given the two binary numbers X = 1010100 and Y =1000011, perform the subtraction (a) $X - Y$; and (b) $Y - X$, by using 2's complement.

(a)

$$
\begin{aligned}
X = &\quad 1010100 \\
\text{2's complement of } Y = &\quad +0111101 \\
\hline
\text{Sum} = &\quad 10010001 \\
\text{Discard end carry } 2^7 = &\quad -10000000 \\
\hline
\text{Answer. } X - Y = &\quad 0010001
\end{aligned}
$$

(b)

$$
\begin{aligned}
Y = &\quad 1000011 \\
\text{2's complement of } X = &\quad +0101100 \\
\hline
\text{Sum} = &\quad 1101111
\end{aligned}
$$

There is no end carry. Therefore, the answer is $Y - X = -$ (2's complement of 1101111) = – 0010001.

Example 3

Repeat Example 2, but this time using 1's complement.

(a) $X - Y = 1010100 - 1000011$

$$
\begin{aligned}
X = \quad & 1010100 \\
\text{1's complement of } Y = \quad & \pm\, 0111100 \\
\text{Sum} = \quad & 10010000 \\
\text{End-around carry} = \quad & \underline{+ \qquad\quad 1} \\
\text{Answer. } X - Y = \quad & 0010001
\end{aligned}
$$

(b) $Y - X = 1000011 - 1010100$

$$
\begin{aligned}
Y = \quad & 1000011 \\
\text{1's complement of } X = \quad & \underline{+\,0101011} \\
\text{Sum} = \quad & 1101110
\end{aligned}
$$

There is no end carry, Therefore, the answer is Y – X = - (1's complement of 1101110) = - 0010001.

Signed Binary Numbers

- In ordinary arithmetic, a negative number is indicated by a minus sign and a positive number by a plus sign.
  - However, computers must represent everything with binary digits.
- It is important to realize that both signed and unsigned binary numbers consist of a string of bits when represented in a computer.
  - The user determines whether the number is signed or unsigned.

Signed Binary Numbers (Cont'd…)

- If the binary number is signed, then the leftmost bit represents the sign, and the rest of the bits represent the number.

  - If the binary number is assumed to be unsigned, then the leftmost bit is the most significant bit of the number

Signed Binary Numbers

Example:

01001 → Unsigned 9

→ Signed +9

11001 → Unsigned 25

→ Signed -9 } Signed magnitude representation

- It is customary to represent the sign with a bit placed in the leftmost position of the number since binary digits.
- The convention is to make the sign bit 0 for positive and 1 for negative

Signed Binary Numbers

- Example:

Signed-magnitude representation:	10001001
Signed-1's-complement representation:	11110110
Signed-2's-complement representation:	11110111

Signed Binary Numbers

Signed Binary Numbers

Decimal	Signed-2's Complement	Signed-1's Complement	Signed Magnitude
+7	0111	0111	0111
+6	0110	0110	0110
+5	0101	0101	0101
+4	0100	0100	0100
+3	0011	0011	0011
+2	0010	0010	0010
+1	0001	0001	0001
+0	0000	0000	0000
−0	—	1111	1000
−1	1111	1110	1001
−2	1110	1101	1010
−3	1101	1100	1011
−4	1100	1011	1100
−5	1011	1010	1101
−6	1010	1001	1110
−7	1001	1000	1111
−8	1000	—	—

Thank you!!!